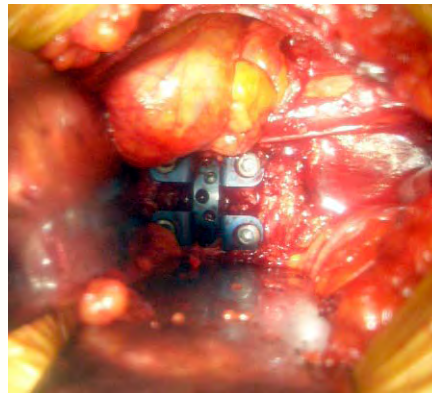
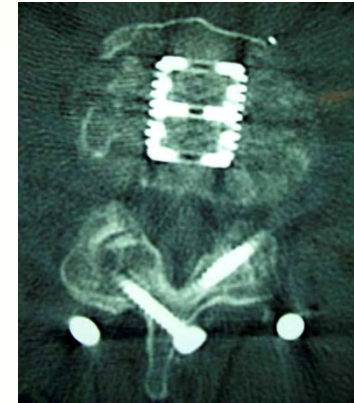
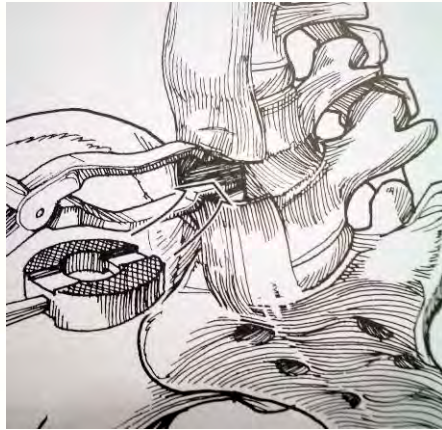
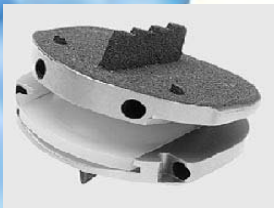


ALIF

(Anterior Lumbar Interbody Fusion)



TDR (Total Disc Replacement)

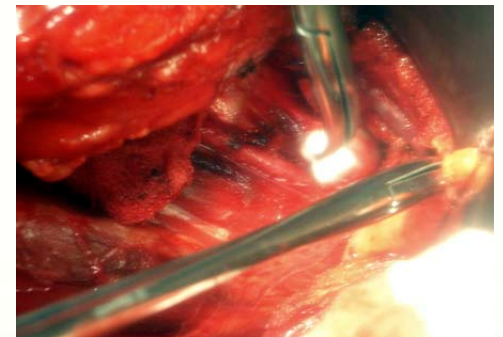


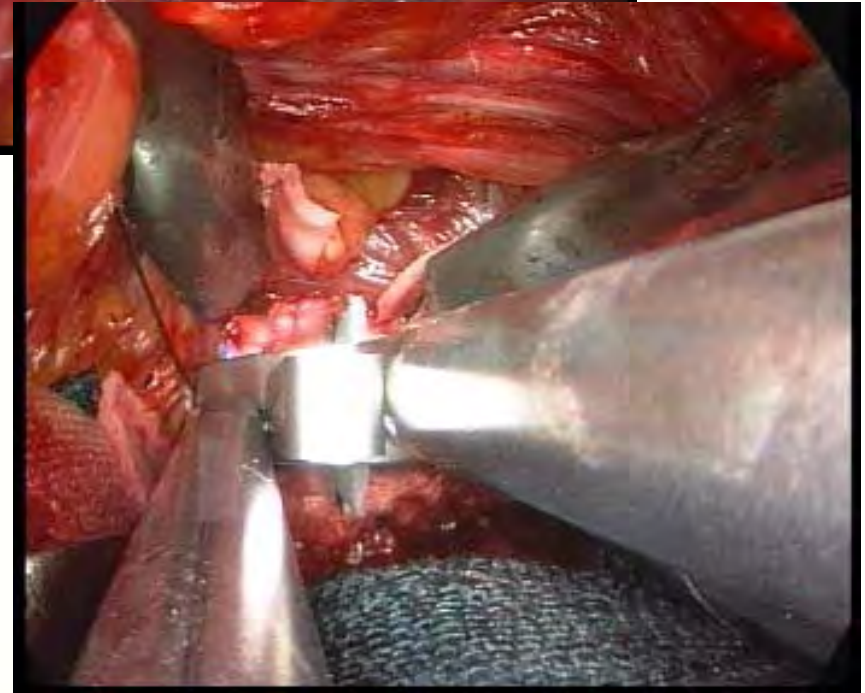
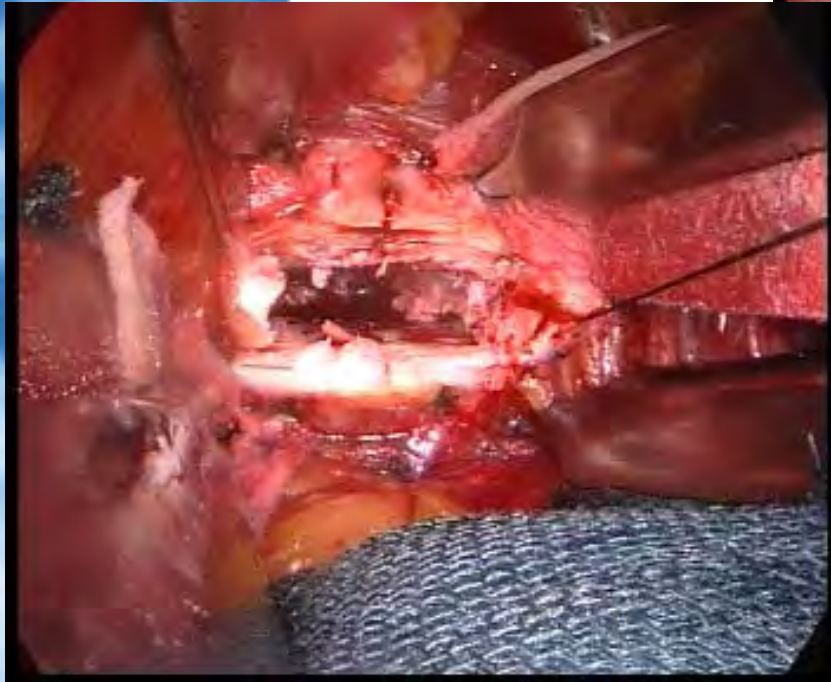
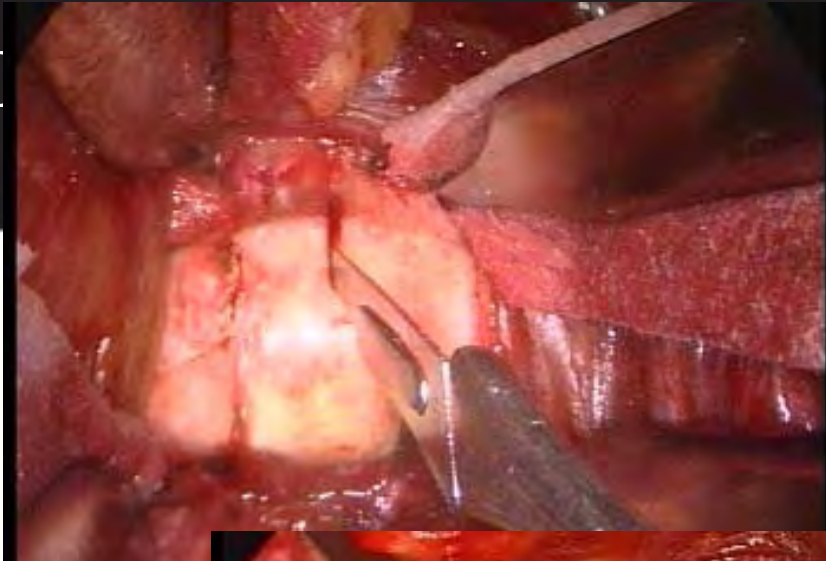
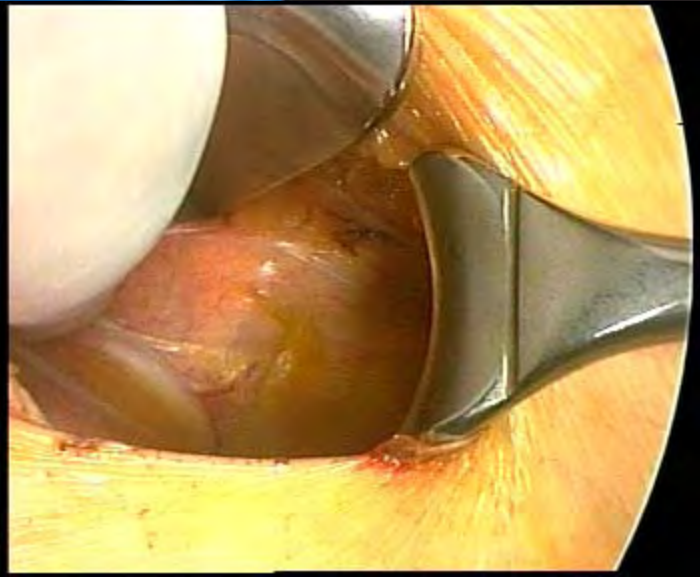
Anterior Approach

Special Considerations Single/ Multilevel TDR



- Indication!!
- Which level?
- How many levels?
- What incision?
- Which route?
- Access surgeon?
- GoreTex Shield?





Summary TDR

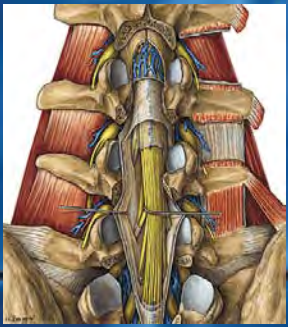
INDICATION IS PARAMOUNT!

SIZE AND TECHNIQUE are IMPORTANT

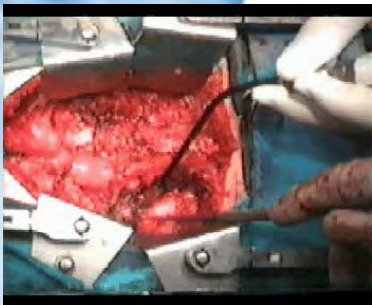
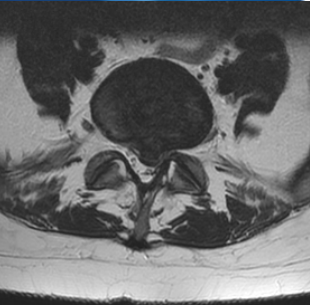


- 1-2 level good outcome
not much different from 1 level
- 3 level rotational (in)stability a concern
preservation of annulus
postOP exercises **absolute MUST**
- 4 level as above
fusion operation of 1st choice
- Hybrids 2+1 3+1

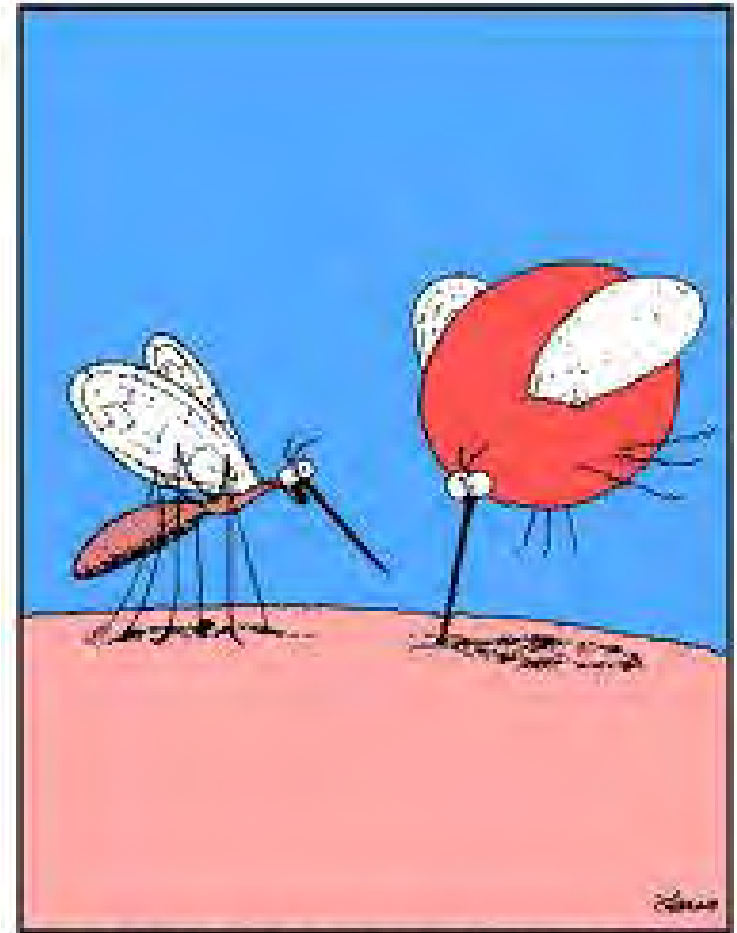
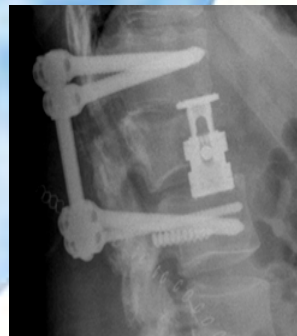
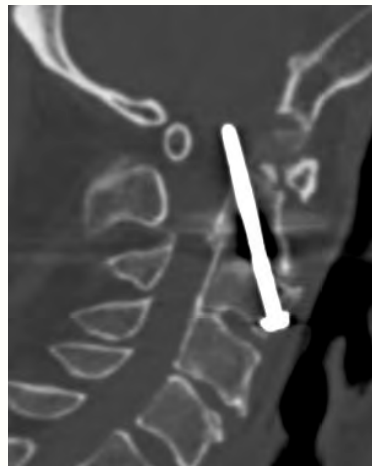
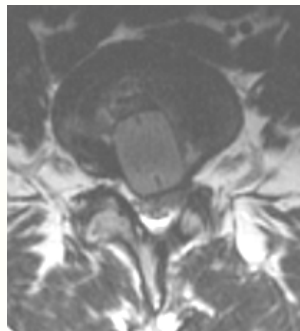
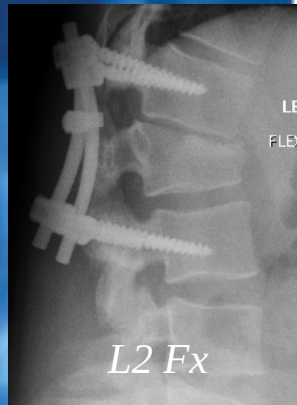
Spine: Postoperative Recovery



- Spinal Unit 3-5-7 days
- Spinal Immobilisation in a brace
- Very light walking
- Slow and gentle movement orientation
- **Aquatic work ASAP as long as wound is healed (ideally 6-8 weeks)**
- Less is more
- **Corrective stretching & soft tissue mobilisation**
- Functional exercises
- **FES Functional electrical stimulator**



Complications



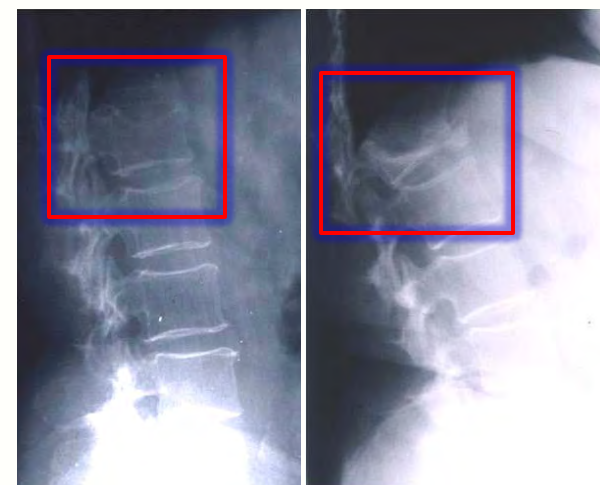
"Pull out, Betty! Pull out! ... You've hit an artery!"

The Osteoporotic Spine

The Concerns

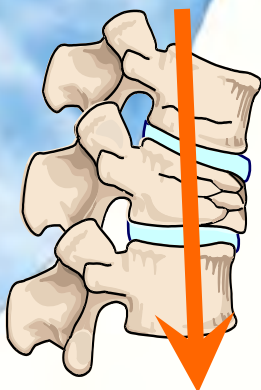


- Progression of **pain**
- Progression of fracture collapse
- Progression of **Kyphosis**
- Diminished **mobility** -> bone loss
- Diminished quality of life
- Diminished **pulmonary function**
- Increased risk of falls
- Increased **subsequent fractures**
- Increased **mortality** (up to 34%)



L1 Fracture

6 weeks later



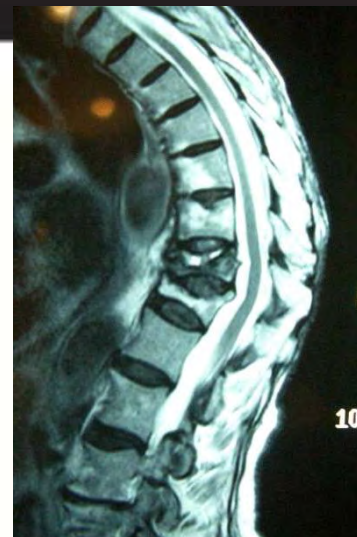
Silverman, *Bone*, 1992; 13:S27-S31
Lyles, *Am J Med*, 1993; 94:596-601
Gold, *Bone*, 1996; 18:185-89

Cooper, *J Bone Mineral Res*, 1992; 13:1001-5
Brauser, Vega, *MedscapeCME*, 2009
Kado, *Arch Intern Med*, 1999; 159:1215-20

The Osteoporotic Spine

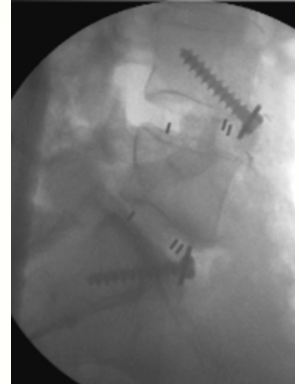
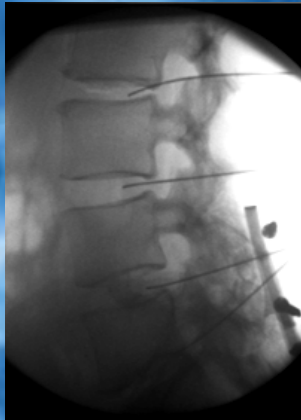
Balloon Kyphoplasty

- ☑ Vertebral body height restoration
- ☑ Significant reduction in pain (95%)
- ☑ Reduced number of days in bed
- ☑ Improved quality of life
- ☑ Improved mobility



FBSS: Conclusion

- 1st OP the best shot the patient / surgeon has
- Communication and Discussion of Indications / Options / Diagnostic Procedures
- Operate the patient and not the images
- Surgical intervention is not the only option
- Early treatment is associated with better outcome



Lee et al., Neurosurg Focus. 2006;21(6)

Spine: Points of Discussion

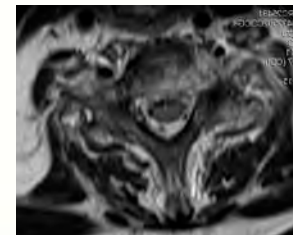
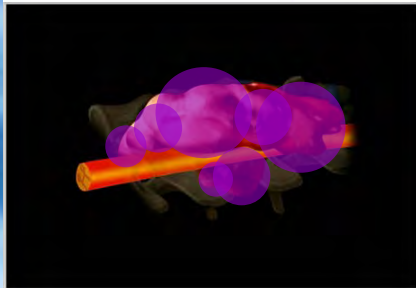
- Should the initial surgical treatment be **MIS**? (pre- and post-FBSS)
- How **aggressive** and **extensive** should the initial surgical treatment be? (pre- and post-FBSS)
- Are we making sure we are meeting patients **expectations** and power of **understanding**?
- Are we using the full potential of dedicated **conservative Tx**?



Tumors & Spine Conclusion



- Curative vs. Palliative / Operoma vs Non-Operoma
- CyberKnife RadioSurgery (LINAC)
- Minimal vs. Complex Intervention, but:
- Stabilization is Paramount



Last but not Least: The Conclusion

*Beware of anybody brandishing the motto:
“We can make you fly...”*





Degenerative Disorders Cervical Spine

- Spondylosis: non-specific term
- Causes
 - imbalance between formation and degradation of proteoglycans and collagen in disc
- Multifactorial
 - Heredity
 - Trauma
 - Environmental (smoking)
 - Metabolic disorders
 - occupational disorder

Degenerative Disorders

Cervical Spine



- **Clinical Presentation**
 - Axial pain - diminished ROM
 - Radiculopathy - FJ-Pain (extension trauma)
 - Cervical myelopathy
- **Radiological Presentation**
 - Disc changes >> FJ changes
 - Mostly C56 and C67
 - No changes if FJ-Pain 2ry hyperextension injury
- **BUT: Asymptomatic patients with +veXR**
 - spondylotic changes 5-95% (age group related)
 - Disc herniation 10% (<40yo) 5% (>40yo)
 - DD changes 25% (<40yo) 60% (>40yo)
 - Stenosis 4% (<40yo) 20% (>40yo)

Degenerative Disorders

Cervical Spine

- **Natural Course of Disease**
 - 79% less pain at 10yrs since onset
 - 43% pain free
 - 50% residual pain
 - 7% severe pain
 - 18% chronic disabling pain
- **Pain level correlates to**
 - severity initial pain level
 - Trauma as initial event
 - Instability /Hypermobility (White criteria >3.5mm slide)
 - **NO** correlation to DD changes

Gore DR: Spine 26, 2001

Degenerative Disorders

Cervical Spine

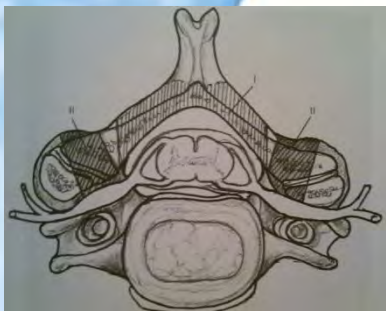
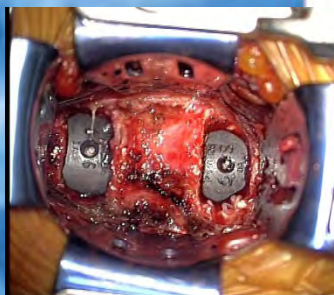
- **Surgical Indications**
 - Instability +/- myelopathy
 - Post-traumatic FJ-injury
 - C1C2 OA
 - (Discogenic neck pain)
 - Provocative discography >> painful>>unreliable
 - 60-70% improvement ACDF+plate
 - Poorer results in >2 levels affected and litigation
 - Cervical disc herniation + neurologic deficits+pain
 - BUT: also consider conservative / injectional Tx
 - Consider Arthroplasty / TDR

Degenerative Disorders

Cervical Spine

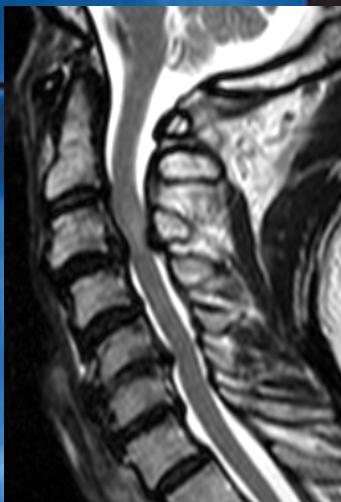
- **Surgical Considerations**

- ACDF +/- plating (**BUT**:fusion rate not related to plating)
 - Pseudarthrosis 5% (1 level) 10-20%(2 level) 30-60% (3)
- Foraminotomy ant./post. (85-90% improved)
- 2 level ACDF vs. Corpectomy
 - Depending on pathology + images
 - Construct biomechanically stronger in 2 level ACDF
- Adjacent Level Degeneration (within 10 yrs ACDF)
 - 20% symptomatic 50% asymptomatic
- Recurrence of Sx in 25% after 20yrs post surgery
- Complications
 - Dysphagia acute 35%, chronic 8-18%
 - Recurrences, Neuro deficit, VA-injury, Hardware 1-2%
 - Graft related collapse/dislodgement 5-25%



Degenerative Disorders

Cervical Myelopathy



- **Clinical**
 - Insidious, dexterity, balance, reflex, PP+Vibratory
 - Pain absent or minimal
- **Causes other than DD changes**
 - OPLL - large central disc herniation
 - Instability - rheumatoid arthritis + C0C1/C1C2
- **Surgical Indication**
 - Most patients with clinical Sx should be treated
 - Non-symptomatic patients need regular work-ups
- **Surgical Considerations**
 - Laminectomy / -plasty >> postOP Kyphosis
 - ACDF+ plate - Laminectomy+posterior fusion



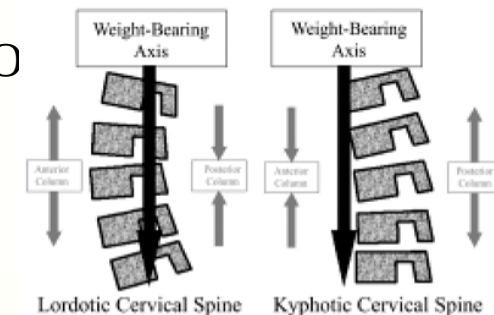
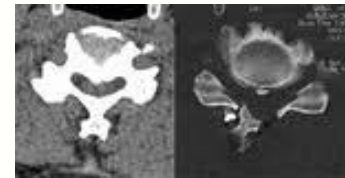
Degenerative Disorders

Cervical Canal Stenosis +/- Myelopathy



- **Surgical Consideration**

- Laminectomy may lead to
 - postOP Kyphosis (25-40%) and Instability
- Laminectomy+posterior fusion
- Laminoplasty
 - 60-75% long-term improvement, BUT:
 - Loss of ROM, recurrent stenosis
 - 30% chronic neck pain
 - 3-5% C5 motor neuropraxia 24-48hrs postO



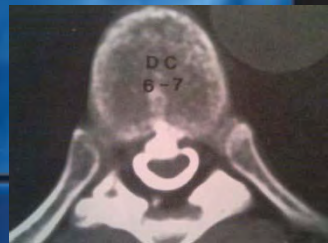
Degenerative Disorders Thoracic Spine



- Same continuum of changes, but less common
- Insidious onset and progress
- Thoracic Disc Herniation
 - Incidence symptomatic 1:1 Million, peak 5th decade
 - Incidence asymptomatic 11-37%
 - commonly lower third, highest occurrence T11T12
 - History of trauma in 25% of cases
- Surgical Considerations depending on
 - Level and location in relation to cord
 - Amount of calcification
 - Number of levels affected
 - Patients age and medical condition



Degenerative Disorders Thoracic Spine



- Surgical Options

- Anterior access

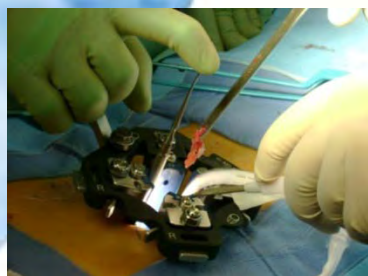
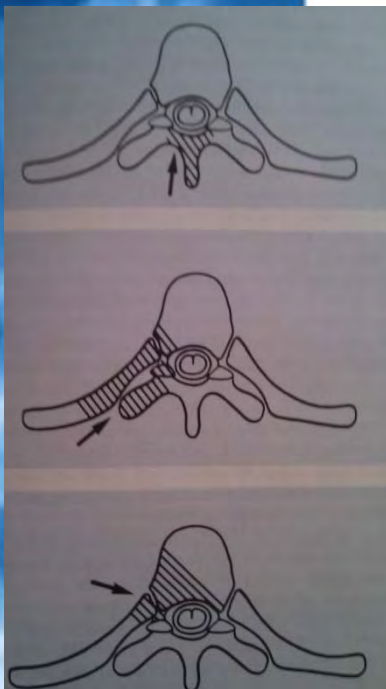
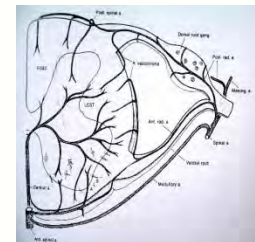
- Upper spine difficult, might require sternal splitting
 - Lower T-Spine, Left-sided thoracotomy (Aorta)
 - T-L-Spine, right-sided retroperitoneal

- CostoTransversectomy

- Standard for draining TB abscesses
 - Risk in vascular watershed area (T4-T9, radicular artery)
 - Resection transverse process and rib (ca. 4cm)

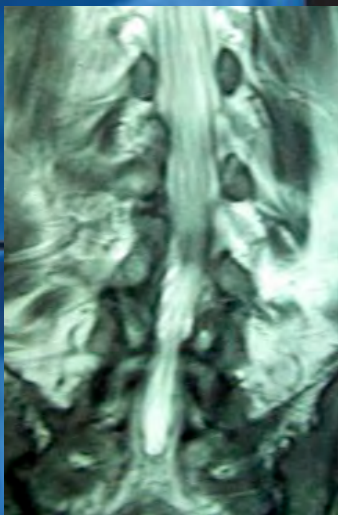
- Surgical Considerations

- Correct level (MR/CT marking, Methylene Blue)
 - Thoracotomy 1-2 levels above affected level
 - Fusion necessary if Sheuerman/ Kyphosis/ VB removal



Degenerative Disorders

Thoracic Stenosis



- Less common than cervical or lumbar stenosis
- Main occurrence lower third of T-Spine and T-L-junction
- Degenerative changes superimposed on preexisting narrowing spinal canal
- Clinical presentation with mixed UMN and LMN Sx
- Associated with L-Spine stenosis
- Anterior vs posterior decompression +/- instrumentation
- Outcome: short-term improvement, long-term deterioration

